



# Simple Models of Complex Slip Distribution? A Case Study of the 2011 $M_w$ 7.1 Van (Eastern Turkey) Earthquake

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**Abstract**—We evaluate the presently proposed simple models for slip distribution, including the homogeneous slip model, the triangular slip model, the  $k$ -square model, the slip tip taper model, and the restricted stochastic source model, to investigate which is most consistent with ‘real-world’ slip distribution inverted. We take the 2011 Van (Eastern Turkey)  $M_w$  7.1 earthquake as an example, considering six inversion results of slip distribution. The Akaike information criterion (AIC) is used to evaluate the models. The evaluation shows that for six inversion results, qualitatively, the  $k$ -square model, with three degrees of freedom, seems most consistent with the real-world slip distribution overall.

**Key words:** Earthquake rupture process, slip distribution, source inversion validation (SIV), Akaike information criterion (AIC), 2011 Van (Eastern Turkey) earthquake.

## 1. Introduction

Whether slip distribution produced by an earthquake rupture, generally heterogeneous and complex, can be described by a simple (sometimes analytical) model with a few degrees of freedom is a fundamental question in seismology, with both theoretical and practical implications. Historically, research addressing this question has been largely limited by the capability of the inversion for earthquake source processes. In recent years, this situation has been dramatically altered with the development of both observation and inversion

algorithms (Ruff 1984, 1987; Ji et al. 2002; Ide 2015). More importantly, accumulation of earthquake cases by several international collaborative projects such as the Source Inversion Validation (SIV) project<sup>1</sup> (Mai and Thingbaijam 2014; Mai et al. 2016b) enables the comparative studies and the exploration of the fundamental issue as per the unified description of slip distribution.

Several models of seismic source process and/or slip distribution have been proposed over the past few decades, including the Haskell model (Haskell 1964, 1966, 1969), the triangular slip model (Manighetti et al. 2005), the slip tip taper model (Scholz and Lawler 2004), and the stochastic source models (e.g., the  $k$ -square model, Herrero and Bernard 1994; Mai and Beroza 2003; Ripperger et al. 2007; Ruiz et al. 2007; Causse et al. 2009; Cultrera et al. 2009; and the restricted stochastic source model, Ward 2004). These models play an important role in the construction of earthquake rupture scenarios, which is important for engineering seismology. However, direct comparisons among these models to determine which is most consistent with the ‘real’ distribution retrieved from observational data is in need of formulation and implementation.

In this article, we take the 2011 Van (Eastern Anatolia, Turkey) earthquake (Altiner et al. 2013; Elliott et al. 2013; Utkucu 2013) as an example to propose a scheme for such comparison. This earthquake occurred on the eastern edge of Lake Van (38.72°N, 43.51°E; Fig. 1), struck on the convergent boundary of the Arabian and the Eurasian plate, and was a typical continental earthquake (Elliott et al. 2013; Utkucu 2013). Several results regarding the source inversion were published after the earthquake

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<sup>1</sup> <http://equake-rc.info/>. Last accessed: August 22, 2018.

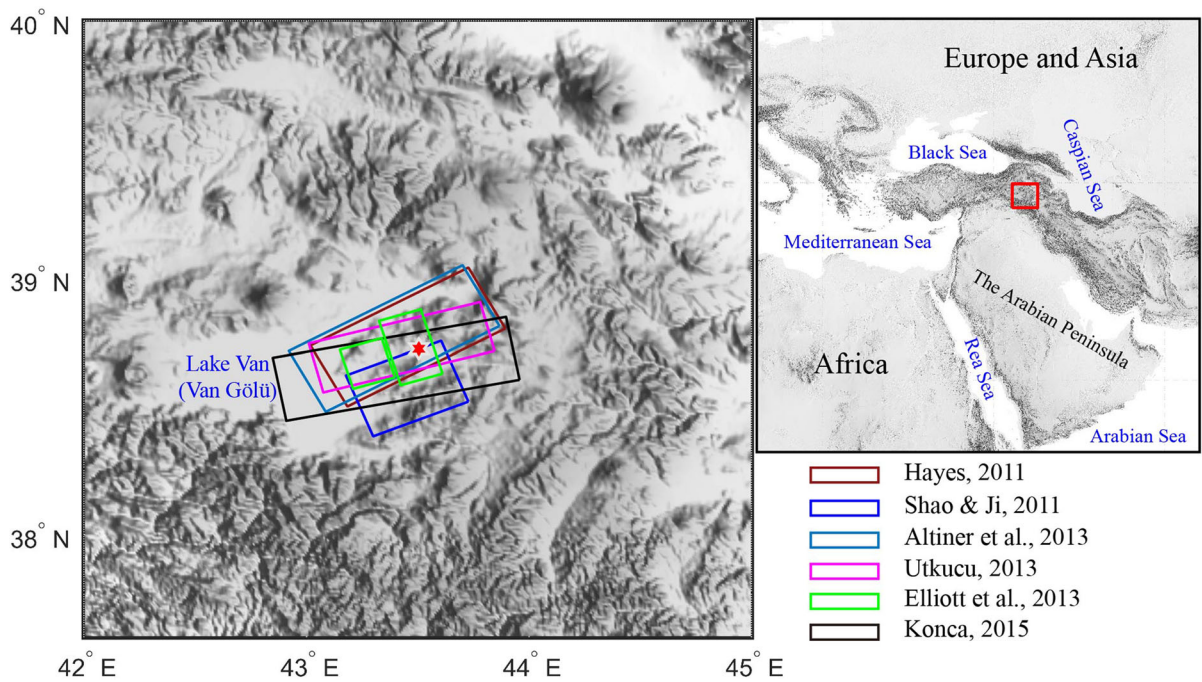


Figure 1

The October 23, 2011  $M_w$  7.1 Van (Eastern Turkey) earthquake, with the surface projection of the inverted static slip distribution. The red hexagon represents the epicenter as released by the United States Geological Survey (USGS). The rectangles with different colors represent the finite-fault seismic source inversion results of different groups, collected and disseminated by the Source Inversion Validation (SIV) Project (<http://equake-rc.info/SRCMOD/searchmodels/allevnts/>, last accessed: August 8, 2018). The indexing figure at the top right shows the region under discussion (37.6°–40.0°N, 42.0°–45.0°E), in which the Arabian Peninsula, Africa, Europe and Asia can be seen from the topography. Background terrain uses the gtopo30 data in the Global Digital Elevation Model (<ftp://edcftp.cr.usgs.gov>, last accessed August 8, 2018)

(Hayes 2011; Shao and Ji 2011; Altiner et al. 2013; Utkucu 2013; Elliott et al. 2013; Konca 2015). We use six inversion results of slip distribution and consider five models for comparison. In the comparison we use the Akaike information criterion (AIC; Akaike 1970, 1974) to evaluate the models.

## 2. Candidate Source Models for Evaluation

For the purpose of ground motion prediction (e.g., in the Uniform California Earthquake Rupture Forecast, version 3, UCERF3<sup>2</sup>), the quantitative analysis of rupture complexity is very important (Somerville et al. 1999; Mai and Beroza, 2003; Causse et al. 2009;

Cultrera et al. 2009). The study must adequately describe the slip heterogeneity to produce realistic scenario earthquakes. The source models describing the rupture complexity can be divided into kinematic and dynamic models, with the former discussed in this article, that is, the dynamic parameters such as the dynamic stress drop  $\Delta\sigma$ ,  $D_c$ , etc., are not considered. Slip distribution in the kinematic models can be roughly divided into two types: deterministic and stochastic. This section briefly introduces five of the proposed source models which will be discussed in this article.

### 2.1. The Homogeneous Model

One of the most widely used source models was the Haskell model (Haskell 1964, 1966, 1969). In this model, a uniform displacement discontinuity

<sup>2</sup> <http://www.wgcepf.org/ucrf3>. Last accessed: August 21, 2018.

propagates within a rectangular fault at a constant rupture velocity. Although the model violates the principles of continuum mechanics, such as the continuity of matter, from a kinetic perspective it is still a very useful first-order approximation to the seismic source process (Ide 2015). Integrating the Haskell model over time produces the homogeneous slip model. This model is a reasonable approximation of a simple seismic rupture propagating along the fault seen by the seismic waves with wavelengths much larger than the fault size (Aki 1968).

## 2.2. The Triangular Slip Model

Manighetti et al. (2005) analyzed 44 surface slip distribution profiles measured after earthquakes and 76 model profiles derived from finite-fault seismic source inversion, and investigated the overall shape of these profiles. The results suggested that these profiles are approximately triangular along both strike and dip, and 70–80% are asymmetric. Long linear slopes and high slip gradients are thus proposed as key ingredients for characterizing earthquake slip profiles.

## 2.3. The $k$ -Square Model

The  $k$ -square model (or the  $k^{-2}$  model) is a stochastic source model for quantifying slip distribution in order to develop characteristic slip models of scenario earthquakes (Mai and Beroza 2003; Burjánek and Zahradník 2007; Gallovič and Brokešová 2007; Ripperger et al. 2007; Ruiz et al. 2007; Causse et al. 2009; Cultrera et al. 2009). Herrero and Bernard (1994) described the static slip with a  $k^{-2}$  asymptotic decay in the wavenumber domain beyond the corner wavenumber  $k_c$ . The corner wavenumber  $k_c$  is inversely proportional to the ruptured fault dimension, and can be viewed as a measure of the roughness of slip distribution (Causse et al. 2010). According to the  $k$ -square model, the slip at wavenumber  $(k_x, k_y)$  is written as

$$D_k(k_x, k_y) = \frac{\bar{D}LW}{\sqrt{1 + \left[ \left( \frac{k_x L}{K} \right)^2 + \left( \frac{k_y W}{K} \right)^2 \right]^2}}, \quad (1)$$

where  $\bar{D}$  and  $K$  are the average slip and the dimensionless constant, respectively, and  $L$  and  $W$  the

length and width of the fault, respectively. Somerville et al. (1999) assumed that  $K$  is equal to 1 (Causse et al. 2010) and proposed a simplified form along the one-dimensional (1D) strike direction:

$$D(k) = \frac{1}{\sqrt{1 + (kL)^4}}. \quad (2)$$

At high wavenumbers, this simplified model has exactly the same decay as the original model proposed by Herrero and Bernard (1994).

## 2.4. The Slip Tip Taper Model

The slip gradients at the tips of an earthquake rupture or fault are usually linear (Scholz and Lawler 2004). The linear fault tip taper (FTT) is a derivative of the constant fault tip taper (CFTT) model, a numerical elastic–plastic crack model in which inelastic deformation occurs in a volume surrounding the crack tip. Scholz and Lawler (2004) determined the FTT by a simple linear fit to the profile in the tip zones, and found that the FTT is scale-independent and increases with the strength of the wall rock. Assuming the middle part of the rupture along the 1D strike or dip direction to be flat, the tip taper model is represented by a trapezoid.

## 2.5. The Restricted Stochastic Source Model

Another version of the stochastic model, the restricted stochastic source model, was proposed by Ward (2004). Assuming that slip tapers to zero at both ends of the rupture, the expected slip at  $l$  along the strike direction is expressed by

$$D(l) = 2\bar{D} \frac{l^q}{L^{2q}} [L - l]^q. \quad (3)$$

For standard Brownian types, taking  $q$  to be 1/2, a set of random variables are superimposed on the normalized model to form a stochastic source model. This model is just one of many stochastic source models and offer several unifying insights into rupture behavior that physics-based simulations do not (Ward 2004).

### 3. Comparison of Different Models

#### 3.1. The Akaike Information Criterion (AIC)

Comparison of different models must balance two factors, namely the match with the real-world data, and the degrees of freedom of the model. For this purpose we use the AIC. Based on information theory, Akaike (1970, 1974) proposed the AIC defined by

$$\text{AIC} = (-2) \log(\hat{\theta}) + 2p, \quad (4)$$

where  $\hat{\theta}$  and  $p$  are the maximum likelihood function of the model and the number of independently adjusted parameters (that is, the degrees of freedom), respectively. According to the principle of parsimony, a model with smaller AIC is ‘better’ than a model with higher AIC.

In this paper, we use the finite-fault seismic source inversion results released by different groups as the ‘real-world’ slip distribution or the ‘ground-truth’. We assume that the model error obeys independent normal distribution, so we use

$$\text{AIC} = n \ln \left( \sum_{i=1}^n (s_{mi} - s_{gi})^2 \right) + 2(p + 1) - n \ln(n), \quad (5)$$

where  $s_{mi}$  and  $s_{gi}$  are the slip of the  $i$ -th sub-fault of a certain model (such as the  $k$ -square model) and the ‘ground-truth’ (such as one of the inversion results), respectively, with  $n$  being the number of sub-faults.

#### 3.2. Degrees of Freedom and Fitting

The uniform fault geometry is used in the five models for comparison; therefore, the geometric parameters of the fault (strike, dip, and rake) are not considered as independent parameters. For the homogeneous slip, the model is constructed as a rectangular rupture surface with uniform average slip ( $\bar{D}$ ), and thus there is one degree of freedom.

The remaining four models are fitted in the 1D direction along the strike. The triangular slip model and the slip tip taper model are similar to the restricted stochastic source model proposed by Ward (2004) in that the slip tapers to zero at both ends of the rupture. A simple global search method is thus

used to find the vertex with the minimum residual fitting the ‘ground-truth’ from 0 to  $L$  for  $l$  and 0 to  $D_{\max}$  for  $D$ . Based on this consideration, the triangle slip model has two degrees of freedom (with two independent parameters,  $l$  and  $D_{\max}$ ). The slip tip taper model, represented by a trapezoid, can also be searched globally on 0 to  $L$  for the abscissas of two upper vertexes ( $l_1, l_2$ ) of the trapezoid with minimal residuals, and thus has three degrees of freedom (with three independent parameters,  $l_1, l_2$ , and  $\bar{D}$ ). Referring to Eq. (3), the restricted stochastic source model has four degrees of freedom (with  $L, q, \bar{D}$  and another degree of freedom describing random numbers fitting the chi-square distribution). We carried out 1000 simulations of the restricted stochastic source model and calculated the AIC with their average residual values.

Considering that the main feature of the  $k$ -square model is characterized in wavenumber domain, we fitted the finite-fault seismic source inversion results according to the simplified form (2) proposed by Somerville et al. (1999) through the Fourier transform. From Parseval’s theorem, the sum of squares of residues in the wavenumber domain is equal to that in the real-world spatial domain. In other words, regardless of whether in the spatial or wavenumber domain, the calculated AIC value for models is the same. The  $k$ -square model has three degrees of freedom in this case, with the three independent parameters  $K, L/W$  and  $\bar{D}$ .

### 4. Case Study: The 2011 Van (Eastern Turkey) Earthquake

#### 4.1. The 2011 Van (Eastern Turkey) Earthquake

The Van (Eastern Anatolia, Turkey) reverse-slip earthquake with  $M_w 7.1$  (USGS) struck north of the city of Van (population 370,000), located on the eastern edge of Lake Van (Fig. 1), on Sunday, October 23, 2011, at 10:41 UTC (local time 13:41). Some agencies successively published the source parameters after the earthquake. According to the USGS, the epicenter was located at 38.72°N, 43.51°E, while the Disasters and Emergency Situations Directorate of Turkey (AFAD) reported

38.68°N, 43.47°E. The magnitude of the earthquake released by different institutions varies from  $M_w 7.1$  to  $M_w 7.3$ , and the reported depth ranges between 5 and 20 km. Up to October 31, 2011, seismic networks have recorded more than 1700 aftershocks with magnitude greater than 2. The earthquake caused devastation near the epicenter. According to the report by the AFAD, the earthquake disaster killed 604 people and caused the collapse of 2000 to 3000 buildings. The tectonic setting around the epicenter is characterized by thrusting in the Zagros fold and thrust belt, and strike-slip faulting along the Bitlis suture zone. This region is the convergent boundary of the Arabian and Eurasian plates, with a slip rate of about 23–26 mm/year (Elliott et al. 2013; Utkucu 2013).

SIV collects six finite-fault seismic source inversion results released for this earthquake, obtained by different groups, which facilitates the comparison. Another reason for choosing this earthquake for model evaluation is its size: If an earthquake were too small, then the inversion results of the slip distribution would be unreliable. If an earthquake were too large, then more complex models (generally composed of several simple models as its ‘sub-faults’) would have to be considered.

#### 4.2. The Source Process

After an earthquake, using seismic, strong motion, geodetic data, and/or tsunami recordings, the seismic source rupture process can be retrieved by source inversion. As the far-field approximation, the representation theorem describes a liner relation between the rupture process  $\Delta u_j(\xi, \tau)$  at  $(\xi, \tau)$  on the fault and the observed displacement  $u_i(x, t)$  at  $(x, t)$ , written as (Mai et al. 2016b; Ide 2015)

$$u_i(x, t) = \int_{-\infty}^{+\infty} d\tau \iint_{\Sigma} \Delta u_j(\xi, \tau) C_{jkpq} G_{ip,q}(x, t; \xi, \tau) v_k(\xi) d\Sigma(\xi), \quad (6)$$

where  $C_{jkpq}$  is the elastic constant matrix,  $v_k(\xi)$  the vector normal to the fault surface  $\Sigma$ , and  $G_{ip}$  the Green’s tensor function, that is, the  $i$ -th component of displacement at  $(x, t)$  due to an impulsive force at

$(\xi, \tau)$  in the  $p$ -th direction. Following the Einstein convention, the subscript  $q$  represents the derivative in the  $\xi_q$  direction. The workflow for finite-fault seismic source inversion includes data selection and pre-processing, choosing and running a proper forward modeling with adequate input parameters, and carrying out the actual inversion (Mai et al. 2016b; Ide 2015). The finite-fault rupture process, no matter static or dynamic, is usually represented by the temporal functions at specific (in most cases, rectangular) sub-fault planes in the source region.

The six results for the Van earthquake collected by SIV (Mai and Thingbaijam 2014; Mai et al. 2016b) are shown in Fig. 1. Following Mai et al. (2016a), we use the inversion results for slip distribution collected in the online SRCMOD database for comparison to evaluate the models. In Fig. 1, the red hexagon is the epicenter released by the USGS. The rectangles with different colors represent the surface projection of the finite-fault seismic source inversion results of Hayes (2011), Shao and Ji (2011), Altiner et al. (2013), Utkucu (2013), Elliott et al. (2013), and Konca (2015), respectively, with the slip distribution shown in Fig. 2. From Figs. 1 and 2, the six inversion results exhibit not only the common characteristics or similarity (the number of asperities and their location and coverage), but also the discrepancy of the slip distribution retrieved by inverting the observational data. The discrepancy lies in two folded reasons: the difference in the inversion algorithms, and the difference in the observational data in use.

Table 1 includes the geometry parameters of the earthquake fault and the sub-fault of the inversion results. In the table,  $dL$  and  $L$  are the length of the sub-fault and the whole fault, respectively, along the strike direction, and  $dW$  and  $W$  the width parameters along the dip direction.  $M_w$  is the moment magnitude, and  $\bar{D}$  the average slip. The last column in the table gives the data used for inversion. Hayes (2011), Shao and Ji (2011) and Utkucu (2013) used teleseismic broadband waveforms, Altiner et al. (2013) and Elliott et al. (2013) used geodetic data, and Konca (2015) conducted the joint-inversion using strong motion, teleseismic broadband waveform, and GPS data.



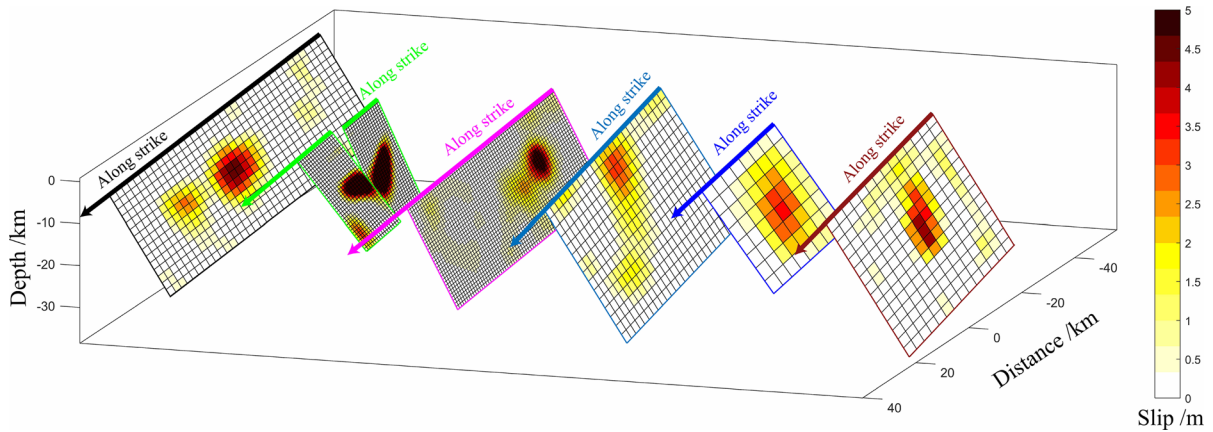


Figure 2

The slip distribution of the 2011  $M_w$  7.1 Van (Eastern Turkey) earthquake. From right to left are the inversion results of Hayes (2011), Shao and Ji (2011), Altiner et al. (2013), Utkucu (2013), Elliott et al. (2013) and Konca (2015), respectively. The colors of the edges of the rectangles correspond to those in Fig. 1. The locations of the inversed epicenters (initial rupture points) is set to be 0 along the strike. The surface is set to be 0 in the direction of depth. Color bar to the right indicates the amount of slip, with a unified scale

Table 1

*Parameters of the finite-fault seismic source inversion results of the 2011 Van (Eastern Turkey) earthquake from different groups*

| No. | Inversion result      | Along strike |          | Along dip |          | $M_w$ | $\bar{D}$ (m) | Data used for the inversion                             |
|-----|-----------------------|--------------|----------|-----------|----------|-------|---------------|---------------------------------------------------------|
|     |                       | $dL$ (km)    | $L$ (km) | $dW$ (km) | $W$ (km) |       |               |                                                         |
| 1   | Hayes (2011)          | 5            | 75       | 3.17      | 50.72    | 7.1   | 0.4           | Teleseismic broadband waveforms                         |
| 2   | Shao and Ji (2011)    | 5            | 45       | 5         | 45       | 7.1   | 0.9           | Teleseismic broadband waveforms                         |
| 3   | Altiner et al. (2013) | 2.96         | 80       | 3.17      | 50.72    | 7.2   | 0.4           | GPS                                                     |
| 4   | Utkucu (2013)         | 1            | 70       | 1         | 35       | 7.1   | 0.6           | Teleseismic broadband body waveforms                    |
| 5   | Elliott et al. (2013) | 1            | 36       | 0.94      | 33.91    | 7.1   | 1.2           | InSAR                                                   |
| 6   | Konca (2015)          | 2.5          | 95       | 2.5       | 40       | 7.1   | 0.5           | Strong-motion, high-rate GPS, teleseismic, and GPS data |

In the table,  $dL$  and  $L$  are the length of the sub-fault and the whole fault, respectively, along the strike direction, and  $dW$  and  $W$  the width along the dip direction.  $M_w$  is the moment magnitude and  $\bar{D}$  the average slip

#### 4.3. Comparison Among the Models

For each of the six inversion results for the 2011 Van earthquake, we compare the five models—homogeneous slip, triangular slip distribution,  $k$ -square, slip tip taper, and restricted stochastic slip—using the AIC value. Such a comparison did not enable us to determine which inversion result was the ‘best’, since this should be evaluated by the fit between the observational data in use and the synthetic data produced by the inversion result. Nor were we able to determine the ‘best’ model, as this requires considering both the power of explanation and the physical basis of the models. What we were

able to obtain is only a relative conclusion, that is, for a certain result of inversion, which simple model seems to have the best ‘fit’, or in other words, which simple model can best ‘describe’ the result of the inversion.

Table 2 shows the degrees of freedom and the associated AIC of the simple source models for each of the six inversion results, with a more visual display in Fig. 3. Figure 3 shows, from left to right, the homogeneous slip, triangular slip,  $k$ -square, slip tip taper, and restricted stochastic source models, with the numbers in brackets after each model indicating the degrees of freedom. The colors of symbols refer to the inversion results corresponding to Figs. 1 and 2

(with index to the top left). It is somewhat interesting to note that although there are significant discrepancies among the six inversion results, as seen from the perspective of the AIC, among the five models, the  $k$ -square model seems to have the best fit. It is noted that triangular slip model performed best for the inversion results of Shao and Ji (2011), with the  $k$ -square model being the second best. From Fig. 3, another feature also seems apparent, that the higher the ‘resolution’ of the inversion (that is, the smaller the sub-faults used in the inversion), the sharper the comparison among different models, as can be seen from the results of Utkucu (2013, pink) and Elliott et al. (2013, green).

### 5. Conclusions and Discussion

Characterization of slip distribution by a simple model with a few degrees of freedom has clear implications for strong motion simulation using scenario earthquake rupture, and may lead to some useful clues regarding earthquake fracture mechanics. In this article, we consider the presently proposed simple models for slip distribution, including the (1) homogeneous slip model, (2) triangular slip model, (3)  $k$ -square model, (4) slip tip taper model, and (5) restricted stochastic source model. We evaluate the fitness of each model to the ‘real-world’ distribution inverted, taking the 2011 Van (Eastern Turkey)  $M_w 7.1$  earthquake as an example. To quantitatively conduct the evaluation, the AIC is used. The result shows that, qualitatively, the  $k$ -square model is most

consistent with the real-world slip distribution, regardless of which inversion result is used for the comparison but the one of Shao and Ji (2011), for which triangular slip model performed best and the  $k$ -square model being the second.

The comparison is implemented thanks to the accumulation of inversion results, especially the data collection efforts by the SIV project. The comparison, however, has limitations due to the current status quo of the source inversion. One problem which needs mentioning is that the discussion of inversion error, which rarely appears in majority of the inversion results. Related to this, the setting of sub-faults in the inversion also has ambiguity. On the other hand, however, it seems that the higher the ‘resolution’ of the inversion (or, the smaller the sub-faults used in the inversion), the sharper the comparison among different models.

Whether one or more parameters are capable of describing the ‘actual’ finite seismic rupture process is still a matter for further discussion. At present, the majority of earthquake catalogs already include the origin time, epicenter location, and magnitude, as well as the strike, dip and rake of the earthquake fault, plus the (half) duration of the source time function. This enables the use of earthquake catalogs with more degrees of freedom, and enriches the description of individual earthquakes and seismic sequences as well. The scaling relations among different parameters of earthquakes may also reveal the physics of earthquakes and faulting (Kanamori and Anderson 1975). It can be expected that if the slip distribution can be well modeled by a few physically meaningful parameters, the earthquake catalog will

Table 2

*The AIC value of seismic source models for different finite-fault seismic source inversions of the 2011  $M_w 7.1$  Van (Eastern Turkey) earthquake*

| Model                 | Degrees of freedom | No. 1<br>Hayes<br>(2011) | No. 2<br>Shao and Ji<br>(2011) | No. 3<br>Altiner et al.<br>(2013) | No. 4<br>Utkucu<br>(2013) | No. 5<br>Elliott et al.<br>(2013) | No. 6<br>Konca<br>(2015) |
|-----------------------|--------------------|--------------------------|--------------------------------|-----------------------------------|---------------------------|-----------------------------------|--------------------------|
| Homogeneous slip      | 1                  | 325.3                    | 115.5                          | 309.7                             | 5054.6                    | 3850.4                            | 767.3                    |
| Triangular slip       | 2                  | 89.3                     | −132.8                         | −159.1                            | 2622.1                    | 4187.1                            | 390.2                    |
| $k$ -square model     | 3                  | −140.0                   | 8.0                            | −747.5                            | −3623.0                   | 850.3                             | −988.7                   |
| Slip tip taper        | 3                  | 1270.6                   | 381.7                          | 2261.3                            | 20,376                    | 12,920                            | 3385.0                   |
| Restricted stochastic | 4                  | 388.1                    | 142.1                          | 456.9                             | 6141.8                    | 4967.5                            | 971.9                    |

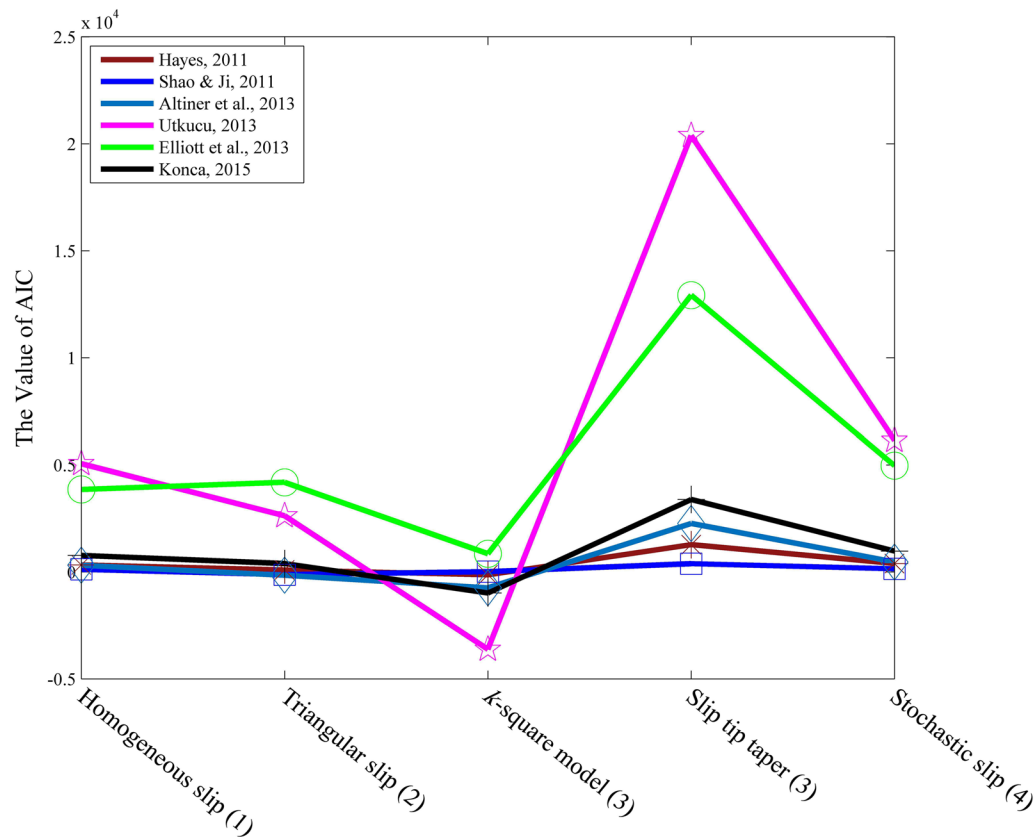


Figure 3

The AIC values of different source models (from the left to the right: the homogeneous slip model, the triangular slip model, the  $k$ -square model, the slip tip taper model, and the restricted stochastic source model, with the numbers in brackets after each model indicating the degrees of freedom), for different finite-fault seismic source inversions of the 2011  $M_w$  7.1 Van (Eastern Turkey) earthquake. The colors of symbols refer to the inversion results corresponding to Figs. 1 and 2 (with index to the top left)

have more information about the nature of earthquakes and have a broader range of application. Toward this end, the present article is just a beginning. The comparison among different models characterizing the time-dependent process of earthquake rupture, as well as its dynamics, still has a long way to go.

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